CHAPTER 8

APPLICATION OF INTEGRALS

CHAPTER ESSENCE AT A GLANCE:
Calculating areas enclosed by curves, lines, parabolas, and ellipses using definite integration with vertical or horizontal strips.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Definite Integral as Planar Area:** The geometric area enclosed between a continuous curve, the horizontal coordinate axis, and boundary vertical lines is computed by evaluating the definite integral of the function across the interval.

• **Exploiting Geometric Symmetry:** Symmetrical figures such as complete circles, ellipses, and centered parabolas can be calculated across a single quadrant and scaled by the appropriate integer symmetry factor.

• **Area Enclosed Between Curves:** The area bounded between two intersecting curves is obtained by integrating the difference between the upper boundary function and the lower boundary function between their intersection coordinates.

• **Alternative Coordinate Orientation:** When boundary curves are naturally expressed as functions of y, integrating horizontal strips between lower and upper horizontal ordinates simplifies algebraic evaluations.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$A = \int_a^b y\,dx = \int_a^b f(x)\,dx$$

Area bounded by curve y = f(x), x-axis, and vertical ordinates

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \rightarrow A = 4\int_0^a \frac{b}{a}\sqrt{a^2 - x^2}\,dx = \pi ab$$

Total enclosed area of ellipse using first-quadrant integration

$$\int \sqrt{a^2 - x^2}\,dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + C$$

Standard circular boundary integration formula

$$A = \int_{x_1}^{x_2} \left[y_{\text{upper}} - y_{\text{lower}}\right] dx$$

Area bounded between two continuous intersecting curves

$$y^2 = 4ax \quad \text{and} \quad x^2 = 4by \rightarrow A = \frac{16ab}{3}$$

Enclosed area between two orthogonal intersecting parabolas

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** [Trap Alert: Missing 4x Multiplier] Full area of ellipse/circle requires multiplying 1st quadrant integral by 4! Forgetting the factor of 4 cuts score in half.

• **Alert:** [Trap Alert: Square Units] Always write 'sq. units' next to the final answer. Evaluators deduct 0.5 Mark if units are missing.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Find the area bounded by curve y = sin x between x = 0 and x = π. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Setup & Integration]** y = sin x between x = 0 and x = π. sin x ≥ 0 on [0, π]. [0.5 Mark]

• Area = $\int_0^{\pi} \sin x \, dx = [- \cos x]_0^{\pi} = - \cos(\pi) - (- \cos 0) = - (-1) - (-1) = 1 + 1 = 2$ sq. units. [1.5 Marks]

• **[Boxed Result: Area = 2 sq. units]**

2. Find the area of the region bounded by the curve $y^2 = 4x$ and the line x = 3 in the first quadrant. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Setup]** Parabola $y^2 = 4x \Rightarrow y = 2\sqrt{x}$. Bounded by x = 0 and x = 3 in first quadrant. [1 Mark]

• **[Integration]** Area = $\int_0^3 y \, dx = \int_0^3 2\sqrt{x} \, dx = 2 [(x^{3/2}) / (3/2)]_0^3 = 2 \cdot (2/3) [3^{3/2}] = (4/3) [3\sqrt{3}] = 4\sqrt{3}$ sq. units. [2 Marks]

• **[Boxed Result: Area = $4\sqrt{3}$ sq. units]**

3. Find the area of the region bounded by $y = x^2$ and the line y = 4. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Setup & Symmetry]** Parabola $y = x^2$ open upwards, intersected by horizontal line y = 4. Symmetric about y-axis: Area = 2 × (Area in first quadrant). Integrating along y-axis from y = 0 to y = 4: $x = \sqrt{y}$. [1 Mark]

• **[Integration]** Area = $2 \int_0^4 \sqrt{y} \, dy = 2 [(y^{3/2}) / (3/2)]_0^4 = (4/3) [4^{3/2}] = (4/3) [8] = 32/3$ sq. units. [2 Marks]

• **[Boxed Result: Area = 32/3 sq. units]**

4. Find the area bounded by the curve $y = x^3$, the x-axis and the ordinates x = -2 and x = 1. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Setup]** $y = x^3$ between x = -2 and x = 1. From x = -2 to 0, y ≤ 0 (below x-axis). From x = 0 to 1, y ≥ 0 (above x-axis). [1 Mark]

• **[Integration]** Area = $| \int_{-2}^0 x^3 \, dx | + \int_0^1 x^3 \, dx = | [x^4/4]_{-2}^0 | + [x^4/4]_0^1 = | 0 - 16/4 | + (1/4 - 0) = |-4| + 1/4 = 4 + 0.25 = 17/4 = 4.25$ sq. units. [2 Marks]

• **[Boxed Result: Area = 17/4 = 4.25 sq. units]**

5. Find the area of the region bounded by parabola $y^2 = 8x$ and line x = 2. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Setup]** Parabola $y^2 = 8x \Rightarrow y = 2\sqrt{2} \sqrt{x}$. Bounded by x = 2 and symmetric across x-axis. [1 Mark]

• **[Integration]** Total Area = $2 \int_0^2 y \, dx = 2 \int_0^2 2\sqrt{2} x^{1/2} \, dx = 4\sqrt{2} [(x^{3/2}) / (3/2)]_0^2 = (8\sqrt{2}/3) [2^{3/2}] = (8\sqrt{2}/3) [2\sqrt{2}] = (8 \times 4)/3 = 32/3$ sq. units. [2 Marks]

• **[Boxed Result: Area = 32/3 sq. units]**

6. Find the area enclosed by the circle $x^2 + y^2 = a^2$ using integration. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Setup & Quadrant Symmetry]** Circle $x^2 + y^2 = a^2$ centered at origin with radius a. By symmetry across four quadrants: Total Area = 4 × (Area in First Quadrant). [1 Mark]

• In first quadrant: $y = \sqrt{a^2 - x^2}$ for x [0, a]. Area = $4 \int_0^a \sqrt{a^2 - x^2} \, dx$. [1 Mark]

• **[Integration using Standard Formula]** = $4 [(x/2) \sqrt{a^2 - x^2} + (a^2/2) \sin^{-1}(x/a)]_0^a = 4 [(0 + (a^2/2) \sin^{-1}(1)) - (0 + 0)] = 4 [(a^2/2)(\pi/2)] = 4 [\pi a^2/4] = \pi a^2$. [2 Marks]

• **[Boxed Result: Area of Circle = πa^2 sq. units]**

7. Find the area of the region bounded by the ellipse x^2 over a^2 + y^2 over b^2 = 1 using integration. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Setup & Formula]** Ellipse x^2 over a^2 + y^2 over b^2 = 1. Total Area = 4 × (Area in First Quadrant) = $4 \int_0^a y\, dx$. From equation: y^2 over b^2 = 1 - x^2 over a^2 = (a^2 - x^2) over a^2 $\Rightarrow$ y = (b over a) $\sqrt{(a^2 - x^2)}$. [1.5 Marks]

• **[Integration]** Area = $4 \int_0^a$ (b over a) $\sqrt{(a^2 - x^2)}\, dx$ = (4b over a) [(x over 2) $\sqrt{(a^2 - x^2)}$ + (a^2 over 2) $\sin^{-1}$(x over a) $]_0^a$ = (4b over a) [(a^2 over 2)(π over 2)] = (4b over a) [π a^2 over 4] = π a b. [2.5 Marks]

• **[Boxed Result: Area of Ellipse = π a b sq. units]**